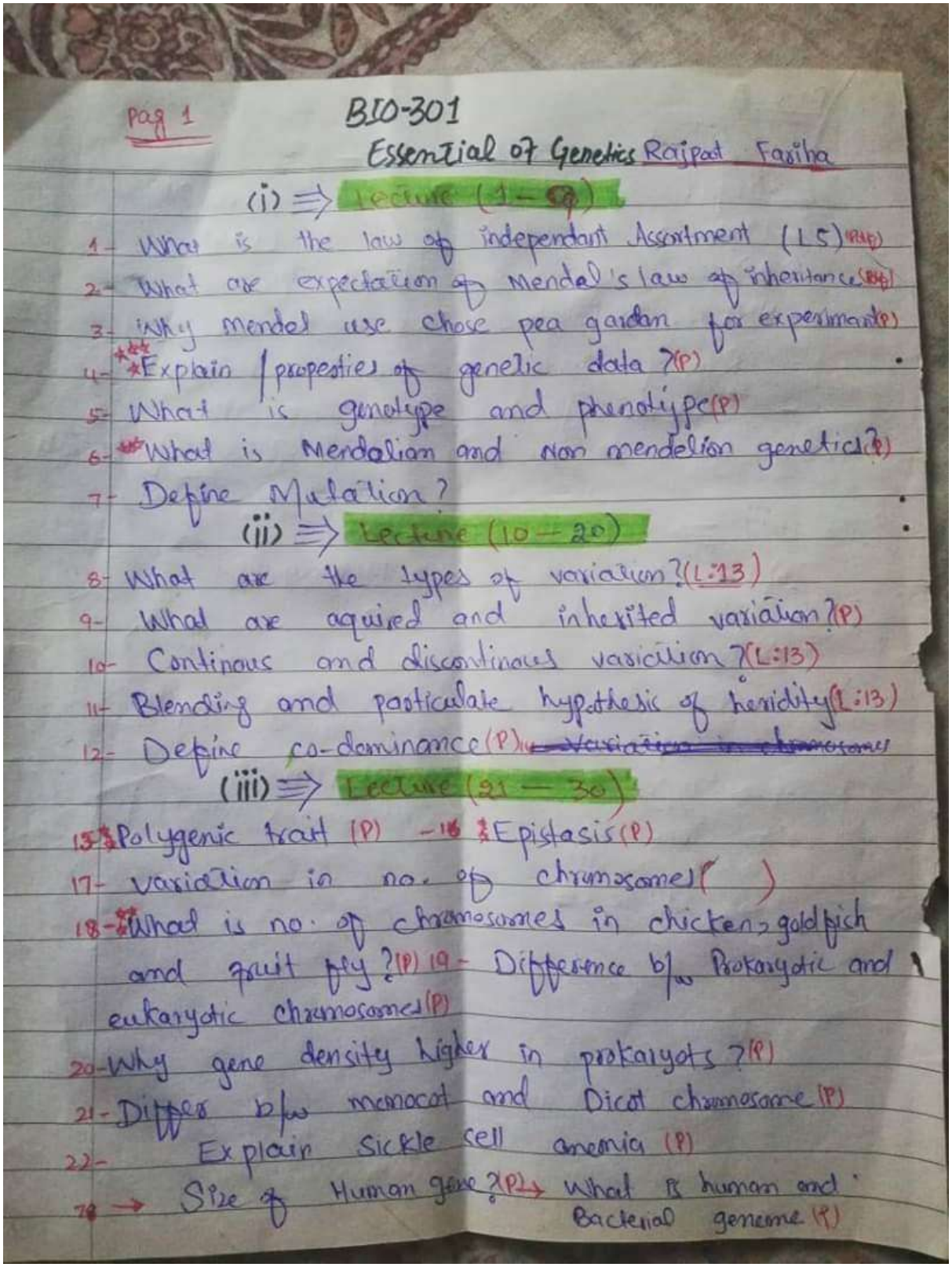
BIO 301 all latest past papers according to lesson sequence



Pag 1

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Essential of Genetics Rajpat Fariha

(i) ⇒ Lecture (1 – 9)

1- What is the law of independant Assortment (L5) (P)(P)

2- What are expectation of Mendel's law of inheritance (P)(P)

3- Why Mendel use chose pea gardan for experimant (P)

4- ***Explain / propesties of genetic data ?(P)

5- What is genetype and phenotype (P)

6- **What is Mendelian and non mendelian genetics (P)

7- Define Mutation ?

(ii) ⇒ Lecture (10 – 20)

8- What are the types of variation ?(L:13)

9- What are aquired and inherited variation ?(P)

10- Continous and discontinous vasicilion ?(L:13)

11- Blending and particulate hypothesis of heridity (L:13)

12- Define co-dominance (P) ~~Variation in chromosome~~

(iii) ⇒ Lecture (21 – 30)

13- *Polygenic trait (P) -14 *Epistasis (P)

17- variation in no. of chromosomes ( )

18- **What is no. of chromosomes in chicken, goldfish and fruit fly ?(P) 19- Difference b/w Prokaryotic and eukaryotic chromosomes (P)

20- Why gene density higher in prokaryots ?(P)

21- Differ b/w monocot and Dicot chromosome (P)

22- Explain Sickle cell anemia (P)

23 → Size of Human gene ?(P) → What is human and Bacterial genome (P)

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(iv) ⇒ Lecture 31 – 40

22- Types of chromosomes (P) 25 ⇒ Linker DNA (Lec 35)
23- Difference b/w Euchromatin and Heterochromatin (P)
24- Write the types of Banding in chromosomes (P)
25- Difference b/w chromosomes and chromatin (P)
26- Define chromosomes and chromatin (P)
27- Types of chromosomes based upon centromere position? (P)
30- Define Kinetochore (P)? 31- What is CEN DNA? (P)
Nucleosome and types of Histone? ⇒ Lecture 40 – 49 (v)
33- Define terms (i) leggiy stend (ii) leading stand (iii) Okazaki fragments → What is GC rule? (P)
34- What is Nucleotides (P) → What is cluster rule? (P)
36 Differ b/w Nucleotide and Nucleoside (L: 44)
37- Name three bonding parts which complete nucleotide (P)
38- What is the length of major and minor grooves in DNA and write no. of nucleoide (P) major groove (P) 39- What Size of human gene (P)

(vi) ⇒ Lecture 50 – 59

40- What is interphase and what are phases of cell cycle (P) (enlist the)
41- Write note on prophase, Metaphase, Anaphase and Telophase (P)
42- Mitotic apparatus (L 59)
43- What is prophase stages of mitosis? (P)

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(vii) ⇒ Lecture 60 – 73

44 – ** Define Spermatogenesis and oogenesis ? (P)
45 – Note of prophase I of miosis (L:64)
46 – ** Define Miosis and Mitosis ? (P)
47 – *** Differentiate b/w miosis and Mitosis (L:73)
48 – Role of Mitosis and Miosis in body ? (P)
49 – ** Describe prophase II of meiosis ? (P)
50 – What do you know about Telophase II of meiosis

(viii) ⇒ Lecture 74 – 84

51 – ** Write the experiment of T.H Morgan (P)
52 – Define Linker map (P)
53 → ** What are SRY gene -Y chromosome. (P)
54 – Name the maker used in linkage map (P)
55 → ** Define genetic mapping (P)
56 → * Types of genetic markers (P)
57 → Gene used in Drosiphila and recombinant frequency among them (P)
58 → Alleleie

[Note:-
Jin question kay Pdf main answer nahi hain. UN kay agy lecture No. mention kar diyan hain]

** = for Repeated important questions

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(ix)

S.Q

⇒ Note
Ya sub Q pdf pr hain per ya kis topic say relate kar ray ya smj nh a rha. So separate likh dijan

59 → Human sex linked disorder (pdf)
60 → Define polygenic traits (pdf)
61 → What is polytene gene with example (pdf)
62 → Define ~~Gene~~ what are three domine of protein? (pdf)
63 → How plasmid as a cloning vector using high and low molecular weight (pdf)
64 → What is ancient pregnancy test ? (pdf)
65 – What is Electrophoresis technique ? (pdf)
66 – Define isopynic conjugation (pdf)
67 – How plasmid play an important role in microbial (pdf)
68 – resistence again antibiotics (pdf)
69 – What is Franklin & wilkin work. (pdf)
70 – Natural plasmid (pdf)
71 – what is green technique (pdf)
72 – Define Proteomic ~~pdf~~ Define Palindromic sequence (pdf)
73 – What is subjonic information (pdf)
74 – VNTR and application of it.
75 → Point Mutation.
76 → Reverse, forward and Neutral mutation.
77 – How Mutant alter the gene function.
80 – Sex linkage:
81 – What is locus chromosom scaffold (pdf)
82 – difference b/w chromosome & chromatid (pdf)
83 – What is frameshift mutation (pdf)
84 – What is fredrick griffith experiment (pdf)

BEST OF LUCK